C

LINEAR ALGEBRA • UNIT 2.01

Systems of Linear Equations & Row Reduction

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 C	Q3 A	Q4 A	Q5 B
Q6 A	Q7 A	Q8 C	Q9 B	Q10 C
Q11 A	Q12 D	Q13 D	Q14 D	Q15 D
Q16 A	Q17 A	Q18 D	Q19 C	Q20 A
Q21 B	Q22 B	Q23 B	Q24 A	Q25 B

Systems of Linear Equations & Row Reduction Practice Exercises

1 MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21

B 14

C 15 ✓

D 225

WORKED SOLUTION & VERIFICATION

$\|\mathbf{u}\| = \sqrt{(-9)^2 + (-12)^2} = \sqrt{225} = 15.$

2 MCQ

A 3×3 symmetric matrix Q_k has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26

B 32

C 29 ✓

D 41

WORKED SOLUTION & VERIFICATION

The trace equals $\lambda_1 + \lambda_2 + \lambda_3 = 29$.

3 MCQ

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

- **A** $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$ ✓
- **B** $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
- **C** $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$
- **D** $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

WORKED SOLUTION & VERIFICATION

To find $T(\mathbf{v})$, we multiply the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Performing the matrix-vector multiplication: $T(\mathbf{v}) = \begin{pmatrix} (2 \times 1) + (1 \times 2) \\ (1 \times 1) + (3 \times 2) \end{pmatrix} = \begin{pmatrix} 2 + 2 \\ 1 + 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$. Thus, the correct answer is $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$.

4 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

- **A** (7, 6) ✓
- **B** (6, 7)
- **C** (8, 5)
- **D** (5, 8)

WORKED SOLUTION & VERIFICATION

The transformation is defined by the matrix multiplication $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$. We are given the point $(3, 1)$, so $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Performing the matrix multiplication:

$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} (2 \times 3) + (1 \times 1) \\ (1 \times 3) + (3 \times 1) \end{pmatrix} = \begin{pmatrix} 6 + 1 \\ 3 + 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$

Therefore, the image of the point $(3, 1)$ is $(7, 6)$.

5 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

- **A** 1
- **B** 0 ✓
- **C** -7
- **D** 25

WORKED SOLUTION & VERIFICATION

The dot product of two vectors $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} c \\ d \end{pmatrix}$ is given by $\mathbf{v}_1 \cdot \mathbf{v}_2 = ac + bd$.

In this case, $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

Therefore, $\mathbf{v}_1 \cdot \mathbf{v}_2 = (3)(-4) + (4)(3) = -12 + 12 = 0$.

A dot product of 0 indicates that the vectors are orthogonal.

6 MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

- **A** 1 ✓
- **B** 2
- **C** 3
- **D** 4

WORKED SOLUTION & VERIFICATION

We are given the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

To transform this matrix into row-echelon form, we can perform the following row operation: $R_2 \leftarrow R_2 - 2R_1$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2-2(1) & -1-2(2) & 0-2(5) \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -5 & -10 \end{array}\right)$$

Now, we can perform another row operation: $R_2 \leftarrow R_2/-5$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \end{array}\right)$$

This matrix is now in row-echelon form. The corresponding system of equations is:

$x + 2y = 5$

$y = 2$

Substitute the value of y into the first equation:

$x + 2(2) = 5$

$x + 4 = 5$

$x = 1$

Therefore, the unique solution is $(1, 2)$, and the value of x is 1.

7 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

- **A** 13 ✓
- **B** 17
- **C** 7
- **D** 169

WORKED SOLUTION & VERIFICATION

The magnitude of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ in R^2 is given by the formula $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$, we have $x = 5$ and $y = -12$. Therefore, $\|\mathbf{v}\| = \sqrt{(5)^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

8 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

B The angle the vector makes with the positive x-axis.

C **The distance of the point (3,4) from the origin.** ✓

D The sum of the components of the vector.

WORKED SOLUTION & VERIFICATION

The length of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ is given by its magnitude, $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. Geometrically, this corresponds to the distance from the origin $(0, 0)$ to the point (x, y) on the Cartesian plane, as described by the Pythagorean theorem. For $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, the length is $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. This value represents the distance of the point $(3, 4)$ from the origin.

9 MCQ

Consider the following system of linear equations represented by an augmented matrix: $\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \end{array}\right]$. What is the value of x in the first equation?

A 4

B **1** ✓

C 2

D 3

WORKED SOLUTION & VERIFICATION

The augmented matrix represents the system of equations: $x + 2y = 5$ and $y = 2$. We are asked to find the value of x. We can use substitution to solve for x. Substitute the value of y from the second equation into the first equation: $x + 2(2) = 5$. This simplifies to $x + 4 = 5$. Subtracting 4 from both sides gives $x = 1$. Therefore, the value of x in the first equation is 1.

10 MCQ

Consider the upper-triangular matrix $W = \begin{pmatrix} 4 & 19 \\ 0 & 8 \end{pmatrix}$. Determine the eigenvalue product $\det(W) = \lambda_1 \lambda_2$.

A 36

B 35

C **32** ✓

D 13

WORKED SOLUTION & VERIFICATION

The eigenvalues are 4 and 8, with product 32.

11 MCQ

Solve for z: $x + y + z = -1$, $y + z = -4$, $z = 1$ (already row-echelon).

- **A** 1 ✓
- **B** 3
- **C** -5
- **D** 2

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = 1$; $y = -4 - (1) = -5$; $x = -1 - (-5) - (1) = 3$.

12 MCQ

Solve for x: $x + y + z = 6$, $y + z = -3$, $z = 2$ (already row-echelon).

- **A** -5
- **B** 2
- **C** 11
- **D** 9 ✓

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = 2$; $y = -3 - (2) = -5$; $x = 6 - (-5) - (2) = 9$.

13 MCQ

Solve for x: $x + y + z = -4$, $y + z = 4$, $z = -7$ (already row-echelon).

- **A** 11
- **B** -7
- **C** -6
- **D** -8 ✓

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = -7$; $y = 4 - (-7) = 11$; $x = -4 - (11) - (-7) = -8$.

14 MCQ

Solve for z: $x + y + z = 4$, $y + z = -3$, $z = -7$ (already row-echelon).

- **A** 7
- **B** 4
- **C** -5
- **D** -7 ✓

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = -7$; $y = -3 - (-7) = 4$; $x = 4 - (4) - (-7) = 7$.

15 MCQ

Solve for y: $x + y + z = 8$, $y + z = 3$, $z = -5$ (already row-echelon).

- **A** 5
- **B** -5
- **C** 10
- **D** 8 ✓

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = -5$; $y = 3 - (-5) = 8$; $x = 8 - (8) - (-5) = 5$.

16 MCQ

Solve for z: $x + y + z = -1$, $y + z = 4$, $z = 7$ (already row-echelon).

- **A** 7 ✓
- **B** 9
- **C** -5
- **D** -3

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = 7$; $y = 4 - (7) = -3$; $x = -1 - (-3) - (7) = -5$.

17 MCQ

Solve for y: $x + y + z = 8$, $y + z = -7$, $z = -3$ (already row-echelon).

- **A** -4 ✓
- **B** 15
- **C** -3
- **D** -2

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = -3$; $y = -7 - (-3) = -4$; $x = 8 - (-4) - (-3) = 15$.

18 MCQ

Solve for z: $x + y + z = 3$, $y + z = -8$, $z = 4$ (already row-echelon).

- **A** 11
- **B** -12
- **C** 6
- **D** 4 ✓

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = 4$; $y = -8 - (4) = -12$; $x = 3 - (-12) - (4) = 11$.

19 MCQ

Solve for z: $x + y + z = -8$, $y + z = 8$, $z = -4$ (already row-echelon).

- **A** -16
- **B** 12
- **C** -4 ✓
- **D** -2

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = -4$; $y = 8 - (-4) = 12$; $x = -8 - (12) - (-4) = -16$.

20 MCQ

Solve for x: $x + y + z = 3$, $y + z = 1$, $z = 5$ (already row-echelon).

- **A** 2 ✓
- **B** 5
- **C** 4
- **D** -4

WORKED SOLUTION & VERIFICATION

Back-substitute: $z = 5$; $y = 1 - (5) = -4$; $x = 3 - (-4) - (5) = 2$.

21 MCQ

Find the root(s) satisfying for y: $x+y+z=3$, $y+z=2$, $z=-5$ (already row-echelon).

- A 1
- B **7** ✓
- C -5
- D 9

WORKED SOLUTION & VERIFICATION

Back-substitute: $z=-5$; $y=2-(-5)=7$; $x=3-(7)-(-5)=1$.

22 MCQ

Find the root(s) satisfying for z: $x+y+z=-2$, $y+z=-3$, $z=1$ (already row-echelon).

- A 3
- B **1** ✓
- C 2
- D -4

WORKED SOLUTION & VERIFICATION

Back-substitute: $z=1$; $y=-3-(1)=-4$; $x=-2-(-4)-(1)=1$.

23 MCQ

Find the root(s) satisfying for z: $x+y+z=2$, $y+z=-1$, $z=3$ (already row-echelon).

- A -4
- B **3** ✓
- C 5
- D 4

WORKED SOLUTION & VERIFICATION

Back-substitute: $z=3$; $y=-1-(3)=-4$; $x=2-(-4)-(3)=3$.

24 MCQ

Solve for y: $x+y+z=-4$, $y+z=4$, $z=5$ (already row-echelon).

- A **-1** ✓
- B -8
- C 5
- D 1

WORKED SOLUTION & VERIFICATION

Back-substitute: $z=5$; $y=4-(5)=-1$; $x=-4-(-1)-(5)=-8$.

25 MCQ

Find the root(s) satisfying for x: $x+y+z=4$, $y+z=2$, $z=6$ (already row-echelon).

- A 4
- B **2** ✓
- C -4
- D 6

WORKED SOLUTION & VERIFICATION

Back-substitute: $z=6$; $y=2-(6)=-4$; $x=4-(-4)-(6)=2$.